Linear Models Lecture 6: Heteroskedasticity

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Mean Independence Does Not Rule Out Heteroskedasticity

Let

$$X \sim \text{Uniform}(-1,1), \qquad Y = X^2 + X\varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0,1)$ and $\varepsilon \perp X$.

The CEF is $m(X) = X^2$, and the CEF error is $e = X\varepsilon$.

Mean independence holds: $\mathbb{E}[e|X] = X\mathbb{E}[\varepsilon|X] = 0$.

But the conditional variance depends on X:

$$\text{Var}(e|X) = X^2.$$

$\mathbb{E}[e|X] = 0$ guarantees that e has no *systematic* pattern with X. It says nothing about the *spread* of e. Heteroskedasticity is the norm, not the exception.

Theory-Based vs. Agnostic Strategies

- Two approaches to handling heteroskedasticity:
  1. **Theory-based (GLS):** Model the variance structure Σ and use it to improve efficiency.
     - Requires knowing (or estimating) *why* some observations are noisier.
     - Example: if the mechanism is proportional growth, take logs—this may eliminate heteroskedasticity at the source (recall the invariance discussion from Lecture 5).
     - If correct, GLS is more efficient than OLS with robust SEs.
  2. **Agnostic (robust SEs):** Use OLS but correct the standard errors for unknown heteroskedasticity.
     - Requires no model for Σ—valid under any form of heteroskedasticity.
     - Sacrifices efficiency for robustness.
- In practice, most applied work uses robust SEs as the default and reserves GLS for settings where theory strongly constrains Σ.

Why homoskedastic SEs can be dangerously wrong (Hansen 4.13)

Under homoskedasticity, we estimate $\hat{\boldsymbol{V}}^0_{\hat{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}s^2$. But the true variance under heteroskedasticity is $\boldsymbol{V}_{\hat{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{D}\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

How far off can this be? To get a scalar we can interpret, consider the simplest case: a single regressor, no intercept ($Y_i = \beta X_i + e_i$), with $\sigma_i^2 = X_i^2$.

The ratio of the true variance to the classical estimate is:

$$\frac{V_{\hat{\beta}}}{\mathbb{E}\left[\hat{V}^0_{\hat{\beta}}\right]} \approx \frac{\mathbb{E}[X^4]}{(\mathbb{E}[X^2])^2}$$

This ratio is the **kurtosis** of X—a measure of how heavy-tailed the regressor distribution is.

- $X \sim N(0, \sigma^2)$: kurtosis $= 3$. Classical SEs off by $\sqrt{3} \approx 1.7\times$.
- Heavy-tailed X (e.g. wages): kurtosis can be 30+. Classical SEs off by $\sqrt{30} \approx 5.5\times$.

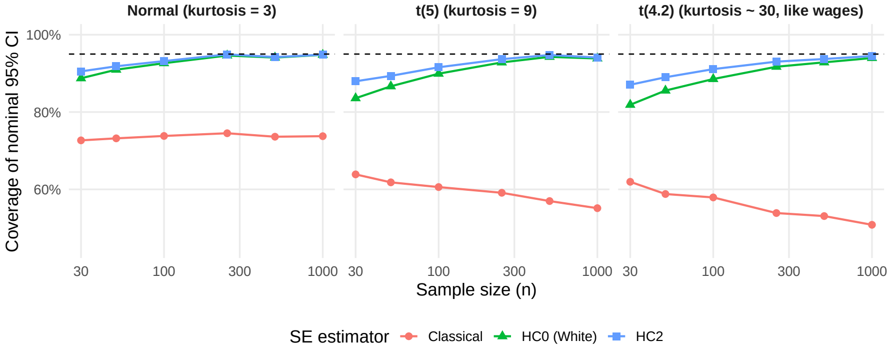
Do Robust SEs Fix the Problem?

- The White (HC0) estimator is **consistent**: it converges to the truth as $n \to \infty$.
- But convergence is *slow* when kurtosis is high—exactly when you need it most.
- HC0 is also biased *downward* in finite samples: it uses $\hat{e}_i^2$, which underestimates σ_i^2 because OLS residuals shrink toward zero (see midterm Q5).
- HC2 corrects this by dividing by $1 - h_{ii}$, but still relies on large-sample approximations.

Question: How bad is the finite-sample performance? We can check by simulation.

Coverage Simulation: Classical vs. Robust SEs

DGP: $Y_i = \beta X_i + e_i$, $\sigma_i = |X_i|$, 5,000 replications. Nominal 95% CI coverage, varying n and regressor kurtosis:



Lessons from the Simulation

- **Classical SEs** dramatically under-cover and *get worse* with heavier tails—more data does not help because the variance formula is wrong, not just imprecise.
- **HC0 (White)** improves with n under mild kurtosis, but with kurtosis ≈ 30 it is still below 80% coverage even at $n = 1{,}000$.
- **HC2** reaches nominal coverage much faster by correcting the finite-sample bias in $\hat{e}_i^2$.

Takeaway: Always use heteroskedasticity-robust SEs—but the version matters. HC2 (or HC3) should be the default, especially with heavy-tailed regressors or $n < 200$.

"Agnostic" Covariance Matrix Estimation under heteroskedasticity

Suppose we have heteroskedasticity:

$$var(\hat{\beta}) = (\boldsymbol{X'X})^{-1}\boldsymbol{X'DX}(\boldsymbol{X'X})^{-1}$$
$$\boldsymbol{D} = \text{diag}(\sigma_1^2, \ldots \sigma_n^2)$$

If we knew $e_1^2, \ldots e_n^2$, then we could just plug them in:

$$\tilde{var}(\hat{\beta}) = (\boldsymbol{X'X})^{-1}(\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' e_i^2)(\boldsymbol{X'X})^{-1}$$

If we use observed $\hat{e}^2$, then this is called the HC0 (heteroskedasticity consistent) estimator, or White covariance matrix estimator.

From the Sandwich to HC Estimators

Recall from Lecture 5: the true variance under heteroskedasticity is the **sandwich**:

$$\boldsymbol{V}_{\hat{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1} \underbrace{\left(\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \sigma_i^2\right)}_{\text{“meat” } \boldsymbol{X}'\boldsymbol{D}\boldsymbol{X}} (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

We never observe $\sigma_i^2 = \text{Var}(e_i|\boldsymbol{X})$. Each HC variant estimates it differently:

Estimator	**Plug-in for σ_i^2**	**Rationale**
HC0	$\hat{e}_i^2$	Simplest; biased downward
HC1	$\frac{n}{n-k}\hat{e}_i^2$	Degrees-of-freedom correction
HC2	$\frac{\hat{e}_i^2}{1-h_{ii}}$	Unbiased: $\mathbb{E}[\hat{e}_i^2|\boldsymbol{X}] = (1-h_{ii})\sigma_i^2$
HC3	$\frac{\hat{e}_i^2}{(1-h_{ii})^2}$	Conservative (jackknife-like)

Problems with White covariance estimators

- HC0 is downward biased in finite samples.
- It is missing the $\frac{1}{n-k}$ term for degrees of freedom.

→ Fixing this gets us HC1.

- It fails to account for leverage:

$$\mathbb{E}[\hat{e}_i^2] = \sigma_i^2(1 - h_{ii}) < \sigma_i^2$$

→ Fixing this gets us HC2

HC2 and HC3: Leverage-corrected estimators

HC2 (unbiased) and HC3 (conservative) correct for the leverage of each observation:

$$\hat{\boldsymbol{V}}_{\hat{\beta}}^{HC2} = (\boldsymbol{X}'\boldsymbol{X})^{-1} \left(\sum_{i=1}^{n} (1 - h_{ii})^{-1} \boldsymbol{x}_i \boldsymbol{x}_i' \hat{e}_i^2 \right) (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

$$\hat{\boldsymbol{V}}_{\hat{\beta}}^{HC3} = (\boldsymbol{X}'\boldsymbol{X})^{-1} \left(\sum_{i=1}^{n} (1 - h_{ii})^{-2} \boldsymbol{x}_i \boldsymbol{x}_i' \hat{e}_i^2 \right) (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Since $(1 - h_{ii})^{-2} > (1 - h_{ii})^{-1} > 1$, we always have:

$$\hat{\boldsymbol{V}}_{\hat{\beta}}^{HC0} < \hat{\boldsymbol{V}}_{\hat{\beta}}^{HC2} < \hat{\boldsymbol{V}}_{\hat{\beta}}^{HC3}$$

HC3 = Prediction Errors in the Sandwich (Lecture 5 Connection)

From Lecture 5: $\mathbb{E}[\hat{e}_i^2|\boldsymbol{X}] = (1-h_{ii})\sigma^2$, so $\hat{e}_i^2$ underestimates σ^2.

HC2 divides once by $(1-h_{ii})$: $\dfrac{\hat{e}_i^2}{1-h_{ii}} \implies \mathbb{E}\left[\dfrac{\hat{e}_i^2}{1-h_{ii}}\middle|\boldsymbol{X}\right] = \sigma^2$ (unbiased)

HC3 divides twice, using the **prediction error** $\tilde{e}_i = \hat{e}_i/(1-h_{ii})$:

$$\frac{\hat{e}_i^2}{(1-h_{ii})^2} = \tilde{e}_i^2 \implies \mathbb{E}[\tilde{e}_i^2|\boldsymbol{X}] = \frac{\sigma^2}{1-h_{ii}} > \sigma^2$$

The second division *deliberately* overestimates σ^2 by $(1-h_{ii})^{-1}$. Why?

> The sandwich meat $\boldsymbol{X'DX}$ is estimated with downward finite-sample bias. HC3 counteracts this by inflating the plug-in variance, especially for high-leverage points where the bias is worst. MacKinnon & White (1985) show HC3 $\approx$ the jackknife variance estimator.

Implementation in R

Option 1: `estimatr` **package** (recommended for applied work)

```
library(estimatr)
# HC2 is the default -- unbiased under homoskedasticity
lm_robust(y ~ x1 + x2, data = dta, se_type = "HC2")
# HC3 is conservative (biased away from zero)
lm_robust(y ~ x1 + x2, data = dta, se_type = "HC3")
```

Option 2: `sandwich` + `lmtest` (flexible, works with any `lm` object)

```
library(sandwich); library(lmtest)
model <- lm(y ~ x1 + x2, data = dta)
coeftest(model, vcov = vcovHC(model, type = "HC2"))
```

R Example: Comparing Standard Errors (Hansen 4.15)

```
library(estimatr); library(sandwich); library(lmtest)
data(mtcars)
m <- lm(mpg ~ wt + hp, data = mtcars)

# Homoskedastic (classical) SEs
se_classical <- summary(m)$coefficients[, "Std. Error"]

# Robust SEs using sandwich
se_hc0 <- sqrt(diag(vcovHC(m, type = "HC0")))
se_hc1 <- sqrt(diag(vcovHC(m, type = "HC1")))
se_hc2 <- sqrt(diag(vcovHC(m, type = "HC2")))
se_hc3 <- sqrt(diag(vcovHC(m, type = "HC3")))

cbind(Classical = se_classical, HC0 = se_hc0,
      HC1 = se_hc1, HC2 = se_hc2, HC3 = se_hc3)
```

Note: HC0 $<$ HC2 $<$ HC3 always holds. The differences grow when observations have high leverage (h_{ii} close to 1).

Simulation: Classical SEs Undercover

```
set.seed(1); B <- 5000; n <- 100; cover_cl <- cover_hc2 <- 0
for (b in 1:B) {
  x <- rexp(n, 1);  e <- rnorm(n, 0, sd = x)  # sigma_i = x_i
  y <- 1 + 2*x + e;  m <- lm(y ~ x)
  ci_cl  <- confint(m)["x",]                        # classical
  ci_hc2 <- coefci(m, vcov.=vcovHC(m,"HC2"))["x",] # robust
  cover_cl  <- cover_cl  + (2 > ci_cl[1]  & 2 < ci_cl[2])
  cover_hc2 <- cover_hc2 + (2 > ci_hc2[1] & 2 < ci_hc2[2])
}
c(Classical = cover_cl/B, HC2 = cover_hc2/B)
# Classical ~ 0.83    HC2 ~ 0.95
```

Classical 95% CIs cover the true β only $\sim$83% of the time. HC2 CIs achieve the nominal 95% coverage.

Notes on Interpretation

- Robust SEs do **not** affect coefficient estimates or R^2/RMSE.
- They only change standard errors, t-statistics, and confidence intervals.
- Use Wald tests (not classical F-tests) for joint hypotheses:

```
library(car)
linearHypothesis(model, c("x1 = 0", "x2 = 0"),
                 white.adjust = "hc2")
```

- Hansen recommends HC2 (unbiased under homoskedasticity) or HC3 (conservative for any $\boldsymbol{X}$) over HC1.
- In most applications HC1, HC2, HC3 are similar. They diverge when some h_{ii} is large (high-leverage observations).

HC1 does not work with Sparse Dummy Variables

- Suppose $Y = \beta_1 D + \beta_2 + e$, where $D_i = 1$ for n_1 cases.
- In the extreme case, $n_1 = 1$:

$$V_{\hat{\beta}} = \sigma^2 (\boldsymbol{X'X})^{-1} = \sigma^2 \begin{pmatrix} 1 & 1 \\ 1 & n \end{pmatrix}^{-1} = \sigma^2 \frac{1}{n-1} \begin{pmatrix} n & -1 \\ -1 & 1 \end{pmatrix}$$

$$V_{\hat{\beta}_1} = \sigma^2 \frac{n}{n-1}$$

- Consider the estimator $\hat{\theta} = \hat{\beta}_1 + \hat{\beta}_2$, with variance $\sigma^2 \frac{n}{n-1} + \sigma^2 \frac{1}{n-1} - \sigma^2 \frac{2}{n-1} = \sigma^2$

$$\hat{V}_{\beta}^{HC1} = s^2 \frac{n}{(n-1)^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\hat{V}_{\theta}^{HC1} = \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{11}} \underbrace{-s^2 \frac{n}{(n-1)^2}}_{V_{12}} \underbrace{-s^2 \frac{n}{(n-1)^2}}_{V_{21}} + \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{22}} = 0$$

Measures of Fit (Hansen 4.18)

- $R^2 = 1 - \frac{\hat{\sigma}^2}{\hat{\sigma}_Y^2}$ *always increases* when regressors are added $\rightarrow$ cannot be used for model selection.
- $\bar{R}^2$ (adjusted) corrects with $(n-1)/(n-k)$, but Hansen argues it still tends to select models with too many parameters.
- **Recommended:** the leave-one-out cross-validation R^2:

$$\tilde{R}^2 = 1 - \frac{\sum_{i=1}^n \tilde{e}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = 1 - \frac{\bar{\sigma}^2}{\hat{\sigma}_Y^2}$$

  where $\tilde{e}_i = \hat{e}_i/(1-h_{ii})$ are the prediction errors.
- $\tilde{R}^2$ estimates the percentage of forecast variance explained; it can be *negative* if the model predicts worse than the mean.
- Hansen: "It is recommended to omit R^2 and $\bar{R}^2$. If a measure of fit is desired, report $\tilde{R}^2$ or $\bar{\sigma}^2$."

R^2 from the Projection Decomposition

Recall $\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ and $\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{P}$ decompose $\boldsymbol{y}$ into orthogonal pieces:

$$\boldsymbol{y} = \underbrace{\boldsymbol{P}\boldsymbol{y}}_{\hat{\boldsymbol{y}}} + \underbrace{\boldsymbol{M}\boldsymbol{y}}_{\hat{\boldsymbol{e}}}, \qquad \hat{\boldsymbol{y}} \perp \hat{\boldsymbol{e}}$$

Since $\hat{\boldsymbol{y}}'\hat{\boldsymbol{e}} = \boldsymbol{y}'\boldsymbol{P}\boldsymbol{M}\boldsymbol{y} = 0$, the Pythagorean theorem gives $\|\boldsymbol{y}\|^2 = \|\hat{\boldsymbol{y}}\|^2 + \|\hat{\boldsymbol{e}}\|^2$, so:

$$R^2 = \frac{\|\hat{\boldsymbol{y}}\|^2}{\|\hat{\boldsymbol{y}}\|^2 + \|\hat{\boldsymbol{e}}\|^2} = 1 - \frac{\|\hat{\boldsymbol{e}}\|^2}{\|\boldsymbol{y}\|^2}$$

This reduces R^2 questions to two facts: (1) $\boldsymbol{P}$ and $\boldsymbol{M}$ are idempotent ($\boldsymbol{P}^2 = \boldsymbol{P}$, $\boldsymbol{M}^2 = \boldsymbol{M}$), and (2) $\boldsymbol{P}\boldsymbol{M} = \boldsymbol{0}$.

Practice: What Happens When You Regress Residuals on $\boldsymbol{X}$?

Suppose a colleague takes your OLS residuals $\hat{\boldsymbol{e}} = \boldsymbol{M}\boldsymbol{y}$ and regresses them on $\boldsymbol{X}$. **Fitted values:** $\boldsymbol{P}\hat{\boldsymbol{e}} = \boldsymbol{P}\boldsymbol{M}\boldsymbol{y} = \boldsymbol{0}$ (since $\boldsymbol{P}\boldsymbol{M} = \boldsymbol{P} - \boldsymbol{P}^2 = \boldsymbol{0}$).

Residuals: $\boldsymbol{M}\hat{\boldsymbol{e}} = \boldsymbol{M}^2\boldsymbol{y} = \boldsymbol{M}\boldsymbol{y} = \hat{\boldsymbol{e}}$ (by idempotency).

R^2**:** The regression explains nothing: $R^2 = \frac{\|\boldsymbol{0}\|^2}{\|\hat{\boldsymbol{e}}\|^2} = 0$.

Key insight: OLS residuals are already orthogonal to $\boldsymbol{X}$. Regressing them on $\boldsymbol{X}$ again produces zero fitted values. This is the content of the normal equations $\boldsymbol{X}'\hat{\boldsymbol{e}} = \boldsymbol{0}$.

Clustered Standard Errors

- Suppose that our data are drawn from groups.

  E.g. students in a school, districts in a state, apartments in a building.
- These observations are subject to common shocks (even if observations don't affect one another).
- These data are called *clustered*.
- We will assume that clusters are known to the researcher and observations are independent across clusters.

Motivating Example: Duflo, Dupas, and Kremer (2011)

In 2005, 140 primary schools in Kenya received funding to hire an extra teacher. Half assigned students to classrooms by prior test score ("tracking").

$$\widehat{TestScore}_{ig} = -0.071 + 0.138\ Tracking_g + \hat{e}_{ig}$$

- With **conventional robust SEs**: $s(\hat{\gamma}) = 0.026$
- With **cluster-robust SEs** (at school level): $s(\hat{\gamma}) = 0.078$

The cluster-robust SEs are $3\times$ larger than the conventional ones!

Ignoring clustering would vastly overstate the precision of the estimated treatment effect. The cluster-robust standard error is the appropriate one because student achievement within a school is correlated.

Formalism for Clustered Regression

- $(Y_{ig},\ \boldsymbol{x}_{ig})$ where $g = 1, \ldots, G$ indexes a cluster and $i = 1, \ldots, n_g$ indexes individuals in cluster g.
- $\boldsymbol{Y}_g = (Y_{1g}, \ldots, Y_{n_g,g})'$, $\boldsymbol{X}_g = (\boldsymbol{x}_{1g}, \ldots, \boldsymbol{x}_{n_g,g})'$ at the group level.

$$
\begin{aligned}
Y_{ig} &= \boldsymbol{x}_{ig}'\beta + e_{ig} \\
\hat{\boldsymbol{\beta}} &= \left(\sum_{g=1}^{G}\sum_{i=1}^{n_g} \boldsymbol{x}_{ig}\boldsymbol{x}_{ig}'\right)^{-1}\left(\sum_{g=1}^{G}\sum_{i=1}^{n_g} \boldsymbol{x}_{ig} Y_{ig}\right) \\
&= \left(\sum_{g=1}^{G} \boldsymbol{X}_g'\boldsymbol{X}_g\right)^{-1}\left(\sum_{g=1}^{G} \boldsymbol{X}_g'\boldsymbol{y}_g\right) \\
&= \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\left(\boldsymbol{X}'\boldsymbol{y}\right)
\end{aligned}
$$

Variance of Clustered Regression

Call $\boldsymbol{\Sigma}_g = E\left[\boldsymbol{e}_g \boldsymbol{e}_g'\right]$ the $n_g \times n_g$ covariance in the g cluster.

$$
\begin{aligned}
var\left[\left(\sum_{g=1}^{G} \boldsymbol{X}_g' \boldsymbol{e}_g\right)\right] &= \sum_{g=1}^{G} var\left[\boldsymbol{X}_g' \boldsymbol{e}_g\right] && \text{(By independence across cluster)} \\
&= \sum_{g=1}^{G} \boldsymbol{X}_g' var\left[\boldsymbol{e}_g\right] \boldsymbol{X}_g \\
&= \sum_{g=1}^{G} \boldsymbol{X}_g' \Sigma_g \boldsymbol{X}_g \\
&\equiv \Omega_n \\
\boldsymbol{V}_{\hat{\beta}} &= \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1} \Omega_n \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}
\end{aligned}
$$

Moulton (1990) Formula

Suppose all clusters are equal size N, homoskedastic within cluster $\mathbb{E}[e_{ig}^2] = \sigma^2$, with intra-cluster correlation $\mathbb{E}[e_{ig}e_{\ell g}] = \sigma^2\rho$ for $i \neq \ell$, and X_{ig} does not vary within a cluster:

$$\boldsymbol{V}_{\hat{\beta}} = \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\sigma^2(1+\rho(N-1))$$

The inflation factor $1+\rho(N-1)$ can be enormous:

ρ	Cluster size N	Inflation factor
0.05	20	1.95
0.10	48	5.7
0.25	48	12.75
0.25	100	25.75

Even modest intra-cluster correlation with moderate cluster sizes produces large distortions in conventional SEs.

Strategy 1: Modeling intra-cluster dependence

- A random-effects model assumes:

$$u_{gi} = \lambda e_g + e_{gi}$$

- Where $e_{gi} \sim f(0,\ \omega^2)$ is individual specific, $e_g \sim f(0,1)$ is cluster wide and the two are independent.

$$\Omega_g = \begin{bmatrix} \lambda^2 + \omega^2 & \lambda^2 & \dots & \lambda^2 \\ \lambda^2 & \lambda^2 + \omega^2 & \dots & \lambda^2 \\ \lambda^2 & \lambda^2 & \dots & \lambda^2 + \omega^2 \end{bmatrix}$$

FGLS

- Estimate pooled model of $\boldsymbol{y}$ on $\boldsymbol{X}$ using OLS. Record $\hat{\sigma}^2$
- Estimate model with fixed effects: regress $\boldsymbol{y}$ on $[\boldsymbol{X}\ \boldsymbol{D}]$. This is our estimate of ω^2, $\hat{\omega}^2$.
- $\hat{\lambda}^2 = \hat{\sigma}^2 - \hat{\omega}^2$.
- Plug into GLS for Σ.
- The Random Effects Estimator is consistent, biased, and asymptotically efficient (if the errors are correct).
- However, if we needed group fixed effects in $\boldsymbol{X}$, we already estimate λe_g, and this model will not work.

Intra-cluster dependence that survives fixed effects

- Contrast the random effects model with a factor model: $u_{gi} = \lambda_{gi} e_g + e_{gi}$, where now λ_{gi} depends on the individual (e.g., some students are affected more by teacher quality than others).
- If we use cluster fixed effects:

$$u_{gi} - \bar{u}_g = (\lambda_{gi} - \bar{\lambda}_g) e_g + e_{gi} - \bar{e}_g$$

$$\text{cov}(u_{gi} - \bar{u}_g, u_{gl} - \bar{u}_g) = (\lambda_{gi} - \bar{\lambda}_g)(\lambda_{gl} - \bar{\lambda}_g)$$

- This is zero only if λ_{gi} is the same for all i — fixed effects remove the common shock but not heterogeneous exposure to it.
- Unfortunately, it is not obvious how to implement this model.

Strategy 2: Cluster Robust Standard Errors Arellano (1987), Hansen (2007)

The squared error e_i^2 is an unbiased estimate for $E[e_i^2]$, $\boldsymbol{e}_g \boldsymbol{e}_g'$ is unbiased for $E[\boldsymbol{e}_g \boldsymbol{e}_g']$. As with White, we can estimate e_i^2 with $\hat{e}_i^2$:

$$\hat{\Omega}_n = \sum_{g=1}^{G} \boldsymbol{X}_g' \hat{\boldsymbol{e}}_g \hat{\boldsymbol{e}}_g' \boldsymbol{X}_g$$

$$\hat{\boldsymbol{V}}_{\hat{\beta}} = a_n \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1} \hat{\Omega}_n \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}$$

$$a_n = \left(\frac{n-1}{n-k}\right) \left(\frac{G}{G-1}\right)$$

a_n is the small sample correction used by STATA, recommended by Hansen (2007).

Clustered SEs in R

Option 1: estimatr (simplest)

```
library(estimatr)
lm_robust(y ~ x1 + x2, data=dta, clusters=cluster_id, se_type="CR2")
```

Option 2: sandwich + lmtest

```
library(sandwich); library(lmtest)
model <- lm(y ~ x1 + x2, data = dta)
coeftest(model, vcov = vcovCL(model, cluster = dta$cluster_id))
```

Option 3: Manual construction (Hansen Ch. 4 R code)

```
xe <- x * rep(e, times = k)          # X_i * e_i
xe_sum <- rowsum(xe, cluster_id)  # sum within clusters
G <- nrow(xe_sum); omega <- t(xe_sum) %*% xe_sum
scale <- G/(G-1) * (n-1)/(n-k)
V_cl <- scale * invx %*% omega %*% invx
```

Inference with Clustered Samples (Hansen 4.22)

- The effective sample size for cluster-robust inference is G (number of clusters), **not** n (number of observations).
- The cluster-robust estimator treats each cluster as a single observation and estimates the covariance from the variation across cluster means.
- If $G = 50$, inference quality is comparable to heteroskedasticity-robust inference with $n = 50$.
- Most cluster-robust theory assumes **homogeneous** cluster sizes. When cluster sizes are highly unequal, cluster sums have heterogeneous variances $\rightarrow$ compounding problem.
- When the number of *treated* clusters is small (e.g. only a few schools received treatment), the cluster-robust SE on the treatment coefficient can be severely downward biased—analogous to the sparse dummy variable problem (Section 4.16).

At what level ought one cluster? (Hansen 4.23)

- Should we cluster by individual, county, state, or region?
- There is a **bias–variance tradeoff**:
  - **Too fine** (e.g. household instead of village): omits covariance terms $\rightarrow$ SEs biased *downward*, spurious significance.
  - **Too coarse** (e.g. state instead of county): adds noise $\rightarrow$ SEs imprecise, less power.
- Rules of thumb:
  - Cluster at the level where treatment is assigned.
  - Cluster at the coarsest level defensible by theory, provided G is not too small ($G \geq 50$ preferred).
  - Your effective sample size is G, not n.
- Honest assessment (Hansen): "We really do not know what is the 'correct' level at which to do cluster-robust inference."

Practical Recommendations (Hansen Ch. 4 Summary)

1. **Always report robust standard errors.** The classical homoskedastic formula $s^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$ is only valid under the strong (and rarely true) assumption of conditional homoskedasticity.
2. **Use HC2 by default** for cross-sectional data. It is unbiased under homoskedasticity and performs well under heteroskedasticity. Use HC3 if you want a conservative alternative.
3. **If data are clustered**, use cluster-robust SEs. Your effective sample size is G (clusters), not n (observations). Cluster at the coarsest defensible level.
4. **For measures of fit**, prefer $\tilde{R}^2$ (leave-one-out cross-validation) over R^2 or $\bar{R}^2$. Hansen: "It is recommended to omit R^2 and $\bar{R}^2$."
5. **Watch for sparse dummies and high leverage.** These can cause robust SE estimators to break down or be severely biased.

Multicollinearity (Hansen 4.20)

- **Strict multicollinearity**: $\boldsymbol{X}'\boldsymbol{X}$ is singular $\rightarrow \hat{\beta}$ is undefined. Typically caused by redundant fixed effects or a dummy variable trap.
- **Near multicollinearity**: regressors are highly correlated. With two regressors and correlation ρ:

$$\text{var}[\hat{\beta}_j|\boldsymbol{X}] = \frac{\sigma^2}{n(1-\rho^2)}$$

  As $\rho \rightarrow 1$, the variance $\rightarrow \infty$.
- This is *not* a bias problem—it is purely a precision problem, equivalent to having a small sample (Goldberger's "micronumerosity" critique).
- Robust standard errors can be sensitive to high leverage under near multicollinearity, producing misleadingly small SEs even when coefficient estimates are imprecise.

Ridge Regression: Trading Bias for Variance

- Near-multicollinearity inflates $\text{Var}(\hat{\beta})$ because $(\boldsymbol{X'X})^{-1}$ has large entries. Ridge adds a penalty that shrinks coefficients toward zero:

$$\hat{\beta}_{ridge} = (\boldsymbol{X}'\boldsymbol{X} + \lambda \boldsymbol{I}_k)^{-1}\boldsymbol{X}'\boldsymbol{y}$$

- This is **biased**: $\mathbb{E}[\hat{\beta}_{ridge}|\boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X} + \lambda \boldsymbol{I}_k)^{-1}\boldsymbol{X}'\boldsymbol{X}\beta \neq \beta$.
- But the variance is smaller: adding $\lambda \boldsymbol{I}_k$ ensures $(\boldsymbol{X'X} + \lambda \boldsymbol{I}_k)$ is always well-conditioned, even if $\boldsymbol{X'X}$ is nearly singular.
- The tradeoff: $\lambda = 0$ gives OLS (unbiased, high variance); large λ gives heavy shrinkage (biased, low variance). Choose λ by cross-validation.
- In R: `glmnet::glmnet(X, y, alpha=0)` or `MASS::lm.ridge`.